Unobserved Is Not Absent:
Graph Neural Network Multiple Imputation for Missing Political Network Data

Updated research proposal and replication of Cranmer, Leifeld, McClurg & Rolfe (2017)

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Abstract

Political network studies almost never observe the whole network, yet most analyze the observed part as if it were complete. This project builds and tests a graph neural network (GNN) method that imputes missing ties and missing actor attributes, produces several completed networks for Rubin's rules, and lets researchers check whether published conclusions survive realistic missingness. Since the proposal, the method has been implemented as an open-source Python package (`relimpute` 0.6.0.dev0), a frozen benchmark of 45 missingness scenarios on 19 political networks has been built and checked. For this class, I replicate and extend Cranmer et al. (2017), who compare logit, MRQAP, latent space (LSM) and exponential random graph (ERGM) models on the Swiss climate policy collaboration network. This document (1) updates the proposal and (2) reproduces the article's Table 1 and Figures 2–5 from the raw replication data. The reproduction succeeds: with the authors' package versions every coefficient of Table 1 is within 0.02 of the published value (logit and MRQAP within 0.005), the BIC values match to within 0.2, and the goodness-of-fit comparison points the same way. The LSM column cannot be reproduced with current software: `latentnet` 2.7.1, which the authors used, transposed asymmetric covariate matrices in directed networks (fixed in version 2.8.0), and I show that this accounts for the whole difference. The other three columns are unaffected. The extension will hide parts of this network under realistic non-response and ask whether the article's substantive conclusions hold when the missing data are handled by deletion, null-tie coding, or GNN multiple imputation.

Replication repository (data, R code, README.txt): https://github.com/kadircihanduran/PLSC-508, folder `Final Project`

1 Updated proposal: where the project stands

The problem. A pair of actors in a network can be observed with a tie, observed without one, unobserved, or structurally impossible. Applied work usually stores the last three as zero. In the four water-policy networks of Metz and Brandenberger (2023), 39 of 149 actors on the roster did not respond (from 10% of actors in Switzerland to 44% in France), and the non-respondents were deleted from the networks together with the ties other actors reported to them. Non-respondents had lower in-degree than respondents in all four countries, so the missingness is not random. In event and archival networks, an unreported cooperation or battle is coded as no tie. Deletion

and null-tie coding bias network statistics and model coefficients once missingness is moderate (Kossinets, 2006; Huisman, 2009; Krause et al., 2020), and principled ERGM-based imputation requires a correctly specified model, converged simulation and small networks.[1]

What has been built since the proposal.

- **Method and software.** The GNN imputer is implemented in the Python package `relimpute` (version 0.6.0.dev0; renamed from `netimpute` to avoid a clash with an unrelated R package of the same name). It records every dyad as tie / no tie / missing / structural, learns from observed ties only, imputes ties and node attributes, and draws m completed networks. For proper multiple imputation, each of the m refits resamples actors and weights each dyad by both actors' bootstrap counts. Results are pooled with Rubin's rules. The package puts 25 imputation models behind one interface: the GNN, plus comparison models such as dyadic regression, latent space, stochastic block model, k-nearest neighbours, network MICE, missForest, Gaussian copula, and ERGM, Bergm and `latentnet` through R. It adds tuning on pseudo-masks of the observed data, a missingness diagnostic with an MCAR permutation test, MNAR sensitivity analysis, and plausibility checks for the imputed networks.
- **Benchmark.** I froze the design before any results were seen. It uses 19 dataset loads (14 cross-sectional networks and 5 four-wave panels, from 16 to 2,733 actors, including the Swiss climate network used below), 45 missingness scenarios with stored masks (random dyads, actor non-response that is MCAR, MAR or MNAR, reconstruction, attribute item non-response, joint unit and item non-response, fixed-choice censoring, panel attrition, false negatives) and about 56 imputation methods. Methods are compared only within the same information tier, with paired differences across 5–20 seeds. Three pilot runs and a code audit found pipeline problems, which have been fixed. The long run (about 630 machine hours) is the next step. *No benchmark results are reported here*, because the pilot runs used one seed and served only to check the code.
- **Paper.** A full draft targeting *Political Analysis* is written up to the results section.

Refined claim and research questions. The pilot runs and the literature check changed the central claim. I no longer expect the GNN to be more *accurate* than the best dyadic or ERGM-based imputer. In single-seed pilot runs it was about on par with a well-specified dyadic model on archival networks and slightly behind it on survey networks. The claim is now that it removes *practical barriers*: no model specification, no MCMC convergence problems, ties and attributes imputed together, and use on networks far larger than ERGM-based imputation can handle. The novelty claim is also narrower, because GNNs have already been used to impute tabular data and node features, and joint imputation of ties and attributes exists without GNNs. To my knowledge, this is the first use of GNNs for multiple imputation of missing ties and actor attributes in survey and archival political networks, benchmarked against ERGM, latent-space, block-model and network-MICE imputation. The questions are:

1. How much do the common fixes (deleting non-respondents, null-tie coding, reconstruction from the other actor's report) change the coefficients and significance of models that political scientists publish?

[1] *Correction to the earlier proposal:* the response rates I reported there for Metz and Brandenberger (2023) (89.2%, 68.4%, 50.5%, 45.5%) do not match the replication data. The actor counts above (non-response 5/52 in CH, 10/39 in DE, 14/32 in FR, 10/26 in NL) come from the replication files, and the earlier statement that non-respondents were kept as nodes was also wrong: they were deleted.

2. Does GNN multiple imputation recover those coefficients, and the tie probabilities, as well as dyadic, latent-space and ERGM-based imputation, especially when missingness depends on network position (MAR and MNAR non-response)?
3. Are its multiple imputations calibrated, so that pooled intervals cover the complete-data estimates at their nominal rate?
4. Under which conditions (size, density, amount and mechanism of missingness) is each method preferable?

Role of the replication in this paper. The replication study is Cranmer et al. (2017). Their Swiss climate network was collected by a roster-based elite survey, which is the setting where actor non-response is the main problem. The network is complete in the replication data, so I can hide parts of it under known mechanisms and compare the results with the complete-data truth. The article also fits the models that applied researchers use downstream (logit, MRQAP, LSM, ERGM) and shows that they can lead to different substantive conclusions on the same data, which makes it a good test bed for the question of how much missing data and its treatment matter for published conclusions. Section 3 describes the extension. Section 2 gives the reproduction.

2 Reproduction of Cranmer, Leifeld, McClurg & Rolfe (2017)

What is reproduced. The data are the directed collaboration network among the 34 organizations central to the Swiss CO_2 act, collected by Ingold (2008) and taken from the authors' Dataverse archive (Cranmer et al., 2016). The article's Table 1 (coefficients and BIC of four models of the collaboration network) and its Figures 2–5 (goodness of fit of each model). The article's other table is a qualitative comparison of the methods, and its online appendix covers further logit variants (Huber–White, GEE, random effects); neither contains estimates of the main models and I do not reproduce them. Figure 1 of the article, which illustrates two ERGM statistics, is also not a result. The four models are the authors': a logit with an indicator for the reciprocal tie; a multiple regression QAP logit with 200 permutations; a two-dimensional Euclidean latent space model with 10,000 posterior draws after a 50,000-iteration burn-in; and an ERGM with 17 terms that include reciprocity, degree and shared-partner statistics. The seed is 12345.

Computational setup. All results are estimated from the raw CSV files, with no saved model object. The authors used R 3.2.4 with `ergm` 3.5.1 and `latentnet` 2.7.1. I ran everything on my Mac (Apple Silicon, R 4.5.2) with `ergm` 3.8.0 and `latentnet` 2.7.1, installed from source. To build under R 4.5, `ergm` needed the memory macros `Calloc/Free/Realloc` renamed to their `R_` forms, and `latentnet` needed two compiler flags, two scalar `&&` conditions that current R rejects, and linking without the Fortran runtime, since it has no Fortran code. No estimator was changed. The script `compat/install_authors_versions.sh` installs these versions into a separate folder, and `AUTHORS_VERSIONS=1` switches the script to them. The script writes the `ergm` terms in a form that runs under old and new versions (for example `odegree1.5`, the new name of `odegreepopularity`).

The article's Table 1. Table 1 is the reproduced Table 1 and Table 2 compares it with the published values. The logit and MRQAP columns equal the published values to two decimals (largest difference 0.005, which is rounding of the published values). The ERGM estimates differ from the published ones by at most 0.007, and the LSM posterior means by at most 0.018, which is Monte Carlo noise of the two samplers. The BIC values of the logit and MRQAP are identical to

the published ones, and those of the LSM and ERGM are within 0.2 (855.73 against 855.89, and 682.94 against 683.03). No LSM interval changes its conclusion about whether zero lies inside it. The published values in Table 2 were transcribed from the article's Table 1; the BIC and coefficient agreement for the three models that do not depend on a sampler (logit, MRQAP) is exact, which also checks the transcription.

Table 1: Reproduction of Table 1 in Cranmer et al. (2017). Logit and ERGM: estimates with standard errors in parentheses. LSM: posterior means with 95% credible intervals in brackets. MRQAP: estimates only. * marks $p < 0.05$ (QAP p-value for MRQAP) or a 95% interval that excludes zero. Produced by `replicate_Cranmer2017.R`.

Variable	Logit	MRQAP	LSM	ERGM
Intercept/Edges	-4.44*	-4.24*	0.93*	-12.17*
	(0.34)		[0.07; 1.83]	(1.40)
Business vs. NGO	-0.86	-0.87	-1.37*	-1.11*
	(0.46)		[-2.39; -0.42]	(0.51)
Opposition/alliance	1.21*	1.14*	0.00	1.22*
	(0.20)		[-0.40; 0.40]	(0.20)
Preference dissimilarity	-0.07	-0.60	-1.76*	-0.44
	(0.37)		[-2.63; -0.91]	(0.39)
Joint forum participation	0.88*	0.75*	1.51*	0.90*
	(0.27)		[0.86; 2.17]	(0.28)
Influence attribution	1.20*	1.29*	0.08	1.00*
	(0.22)		[-0.39; 0.55]	(0.21)
Alter's influence indegree	0.10*	0.11*	0.01	0.21*
	(0.02)		[-0.03; 0.04]	(0.04)
Influence absolute diff.	-0.03*	-0.06*	0.04	-0.05*
	(0.02)		[-0.01; 0.09]	(0.01)
Alter = Government actor	0.63*	0.68*	-0.46	1.04*
	(0.25)		[-1.06; 0.14]	(0.34)
Ego = Environmental NGO	0.88*	0.99	-0.60	0.79*
	(0.26)		[-1.29; 0.09]	(0.17)
Same actor type	0.74*	1.12*	1.18*	0.99*
	(0.22)		[0.63; 1.71]	(0.23)
Mutuality	1.22*	1.00*		0.81*
	(0.21)			(0.25)
Outdegree popularity				0.95*
				(0.09)
Twopaths				-0.04*
				(0.02)
GWIdegree (2.0)				3.42*
				(1.47)
GWESP (1.0)				0.58*
				(0.16)
GWOdegree (0.5)				8.42*
				(2.11)
BIC	808.72	776.41	855.73	682.94

Table 2: Reproduced against published values: largest absolute difference over each model's coefficients, and BIC.

Model	Coefficients	Largest abs. difference	BIC (reproduced)	BIC (published)
Logit	12	0.005	808.72	808.72
MRQAP	12	0.005	776.41	776.41
LSM	11	0.018	855.73	855.89
ERGM	17	0.007	682.94	683.03

Substantive conclusions. The article's message is that the four tools can lead to different conclusions on the same data, and the reproduction shows the same. (1) The LSM disagrees with the other three models about the two covariates that capture alliances and influence. Its estimates for *Opposition/alliance* (0.00, 95% interval $[-0.40, 0.40]$) and *Influence attribution* (0.08, $[-0.39, 0.55]$) are null, while the logit (1.21 and 1.20), the MRQAP (1.14 and 1.29) and the ERGM (1.22 and 1.00) all find clear positive effects (but see the note on software versions below: this LSM result comes from a software bug). (2) Only the LSM finds a significant effect of preference dissimilarity (-1.76, $[-2.63, -0.91]$); the logit (-0.07, s.e. 0.37), the MRQAP (-0.60) and the ERGM (-0.44, s.e. 0.39) do not. (3) The government-receiver effect is positive and significant in the logit (0.63, s.e. 0.25) and the ERGM (1.04, s.e. 0.34), but negative and not significant in the LSM (-0.46). (4) The ERGM has by far the lowest BIC (682.9), followed by the MRQAP (776.4), the logit (808.7) and the LSM (855.7). That the LSM has the highest BIC is expected, because it has no reciprocity or degree terms, so the endogenous structure that the ERGM models has to be absorbed elsewhere.

Figures 2–5: goodness of fit. Figures 1–5 reproduce the goodness-of-fit assessments of the logit, MRQAP, LSM and ERGM. For each model, 100 networks are simulated and six statistics of the observed network (grey lines in the article, blue here) are compared with their distribution across the simulated networks: dyad-wise and edge-wise shared partners, geodesic distances, in-degree, out-degree and incoming k-stars. The logit and MRQAP simulate independent dyads from the fitted probabilities, the LSM draws from the posterior, and the ERGM simulates from the fitted model with the authors' burn-in (30,000) and interval (10,000). I plotted the same statistics with my own code because the authors' plotting function comes from `btergm`, so the figures look different but show the same comparison. As a summary that the article does not report, I counted how many of the 70 binned statistics of the observed network lie inside the central 95% of the simulated values: 99% for the ERGM (69 of 70), against 79% for the MRQAP, 76% for the logit and 70% for the LSM. The ERGM is the only model that reproduces the observed network's degree distributions and shared-partner patterns, as the article concludes. Figure 4 shows the estimated latent positions that accompany the article's Figure 4.

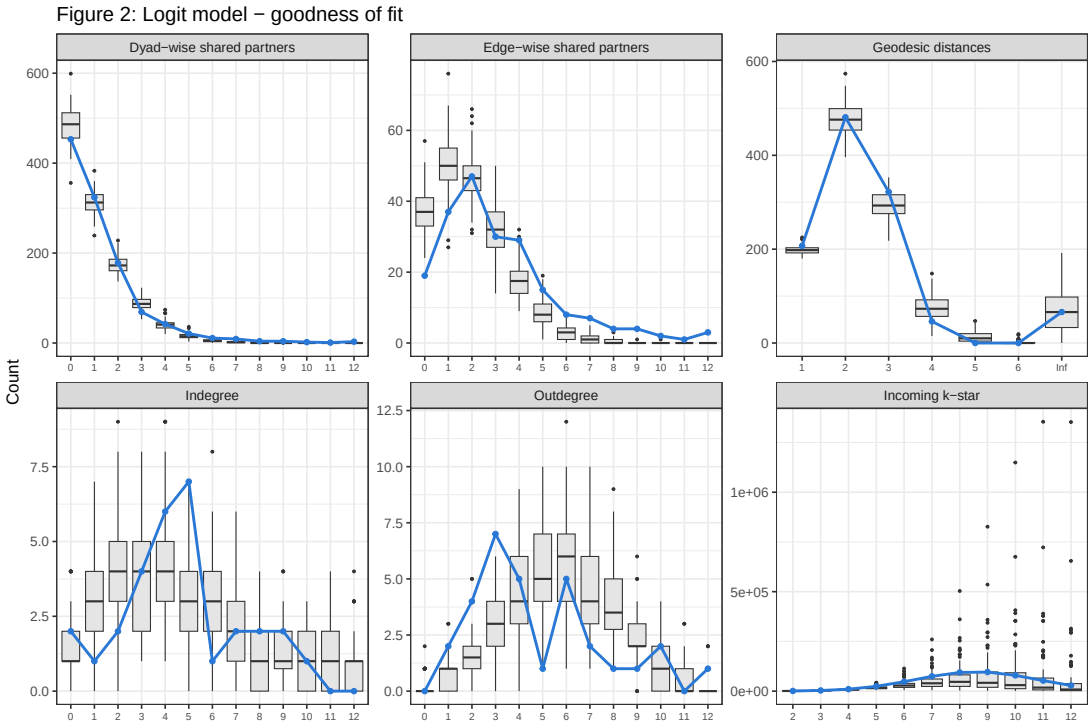


Figure 1: Reproduction of Figure 2 in Cranmer et al. (2017): logit model, goodness of fit. Boxplots: 100 simulated networks. Blue line: observed network.

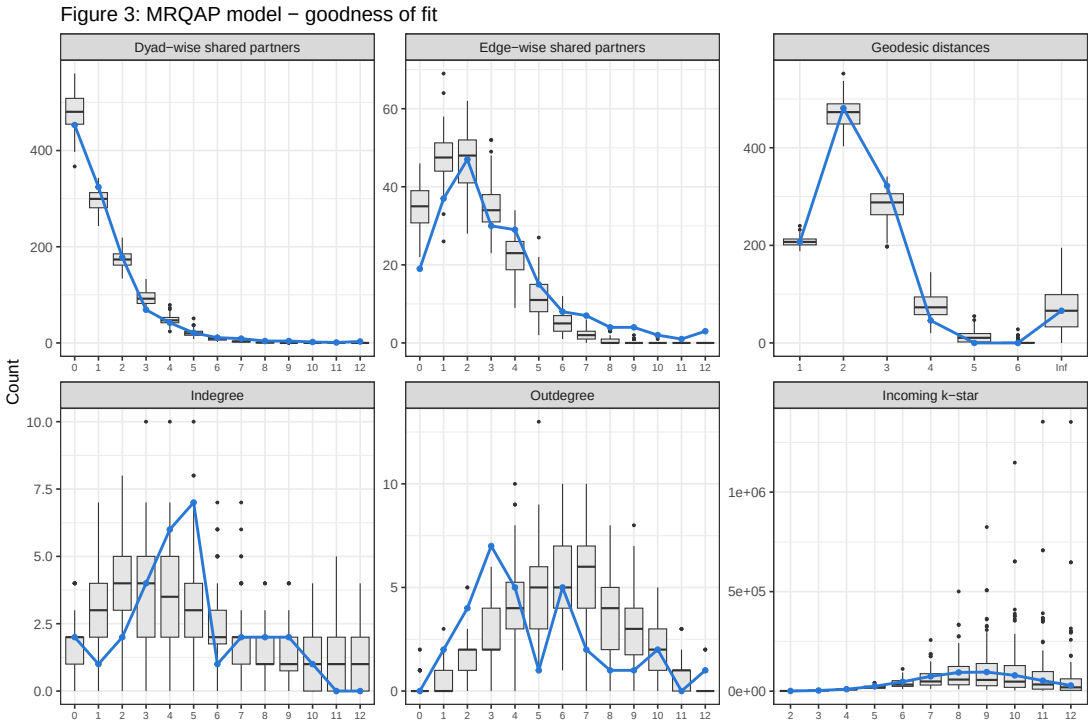


Figure 2: Reproduction of Figure 3: MRQAP model, goodness of fit.

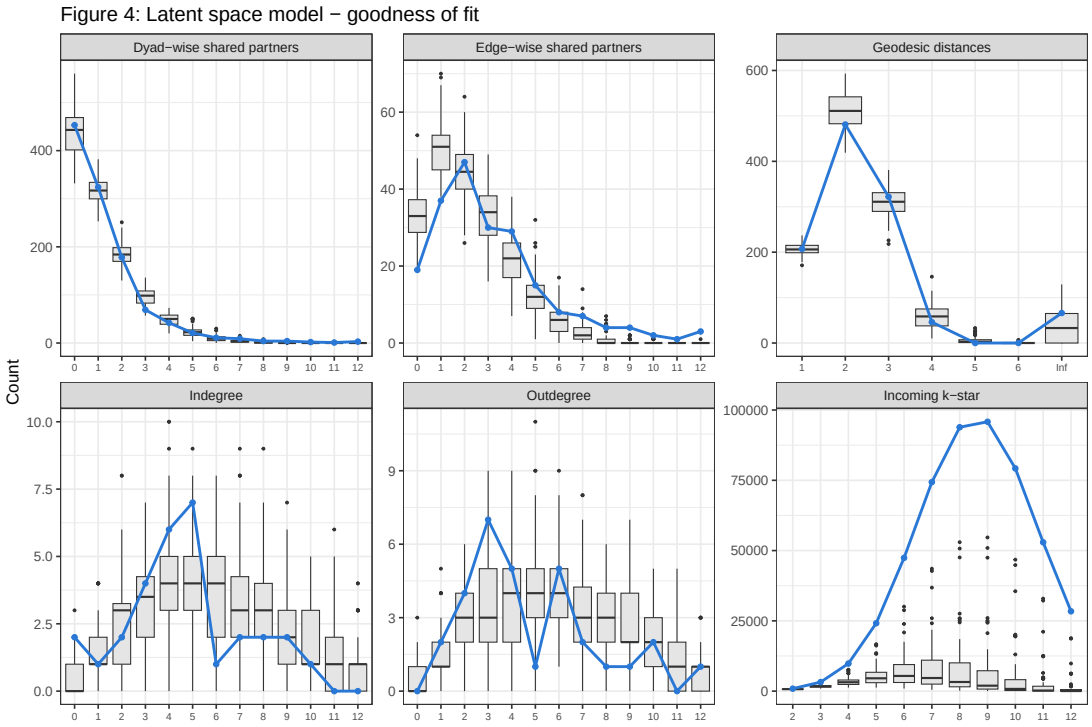


Figure 3: Reproduction of Figure 4 (first part): latent space model, goodness of fit.

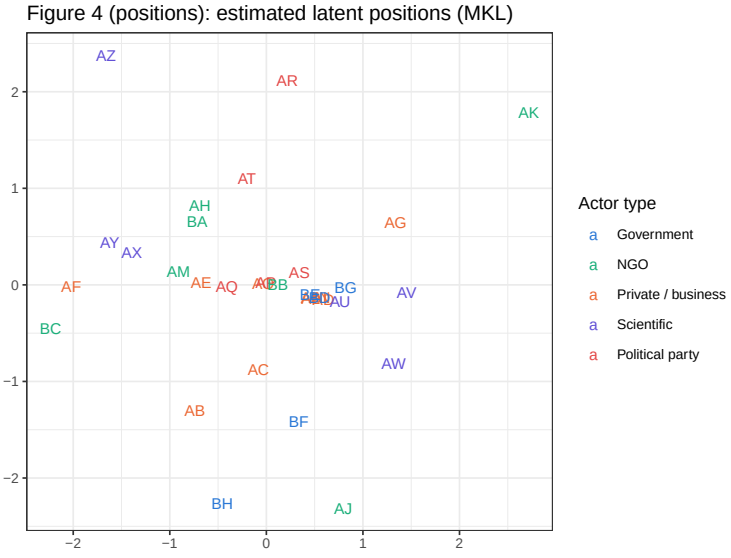


Figure 4: Reproduction of Figure 4 (second part): estimated latent positions of the 34 actors (minimum Kullback–Leibler positions).

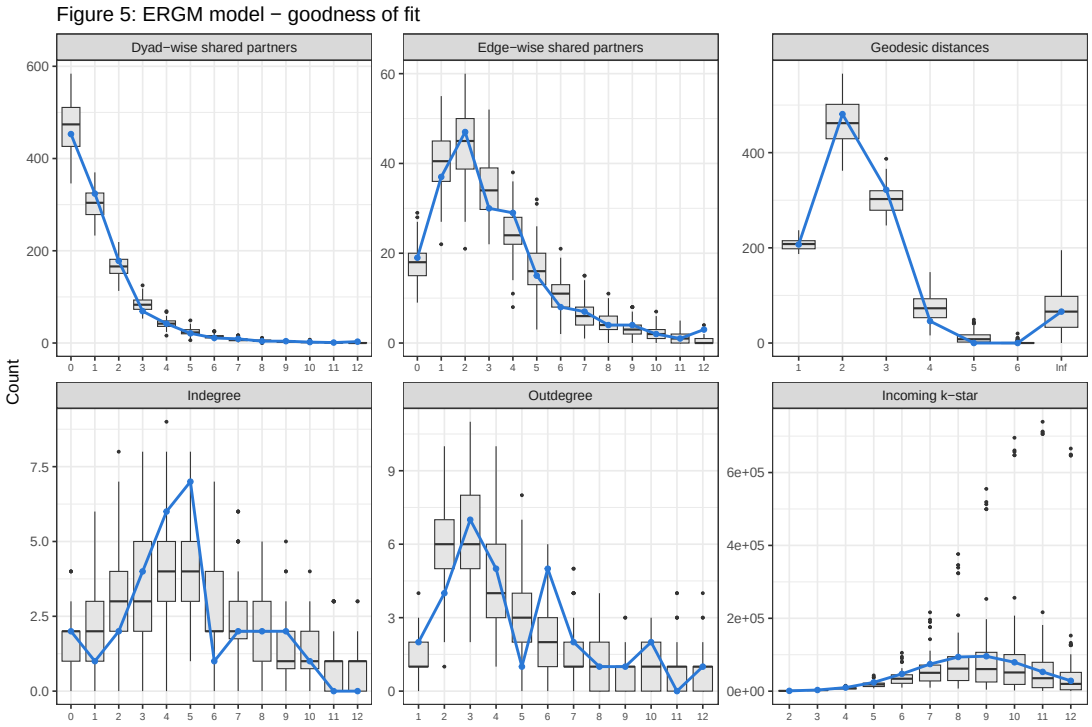


Figure 5: Reproduction of Figure 5: ERGM, goodness of fit.

A note on software versions: why the LSM column cannot be reproduced with current software. I ran the authors' unchanged `analysis.R` on current software (R 4.5.2, `ergm` 4.10.1, `latentnet` 2.12.0). The logit and MRQAP columns are identical to the published ones and the ERGM estimates differ from them by at most 0.021. The LSM changes completely: the intercept is -2.00 instead of 0.95, *Opposition/alliance* is 1.46 instead of 0.00, *Influence attribution* is 1.61 instead of 0.08, *Alter = Government actor* is 0.71 instead of -0.46, *Ego = environmental NGO* is 1.94 instead of -0.60, and the BIC is 918.0 instead of 855.9. The cause is a bug in `latentnet` 2.7.1, the version the authors used. The release notes of version 2.8.0 (August 2017) report that the sampler transposed asymmetric covariate matrices (`edgecov`, `nodeicov`, `nodeocov`) in models for directed networks. Five of the LSM's covariates are asymmetric: the government-receiver and NGO-sender indicators, influence attribution, alliance/opposition and the receiver's influence reputation. I checked this directly. In `latentnet` 2.7.1, fitting the same model to the transposed matrices (and `nodeocov` in place of `nodeicov`) reproduces every coefficient and credible interval of the current-software fit to within 0.01 (for example intercept -1.99 with interval $[-2.88, -1.10]$; *Opposition/alliance* 1.46 with $[0.99, 1.95]$; *Ego = environmental NGO* 1.94 with $[1.06, 2.87]$). So the published LSM column was estimated with those covariates transposed. Its *Alter = Government actor* and *Ego = Environmental NGO* effects, for example, in effect described the ego and the alter the other way round, and its null effects of alliance and influence are artefacts of the bug. For the model as specified, the current-software estimates (alliance 1.46, influence attribution 1.61) agree in sign and significance with the logit, MRQAP and ERGM. The numbers in Table 1 are the published ones, reproduced with `latentnet` 2.7.1; I report them for that reason. The same bug affects the LSM column of Minhas et al. (2019), which uses the same specification and package version.

3 Planned extension: does the comparison survive missing data?

The article treats the Swiss network as complete. Elite policy surveys rarely are: in comparable surveys 10–44% of rostered actors do not respond (Metz and Brandenberger, 2023). The extension asks whether the article's conclusions hold under realistic non-response. The first is that the ERGM fits the network far better than the logit, MRQAP and LSM. The second is that the four tools disagree about which covariates matter, and the question is which of these disagreements are real and which are artefacts of missing data. The steps are:

1. **Hide data with the frozen benchmark scenarios.** I use the same masks that the `relimpute` benchmark stores for this network. Actor non-response hides 10, 25 or 40% of actors' outgoing ties, at random (MCAR), more often for actors with low observed in-degree (MAR, the pattern found by Metz and Brandenberger), or more often for peripheral actors (MNAR). Further masks cover fixed-choice censoring (each actor names at most 3, 5 or 10 partners) and random dyad loss, each with 20 seeds.
2. **Treat the missing data in four ways:** (a) delete non-respondents, as Metz and Brandenberger (2023) did; (b) code unobserved ties as zero; (c) reconstruct from the other actor's report; and (d) multiple imputation with $m = 20$ completed networks from the `relimpute` GNN, with dyadic and latent-space imputers as baselines.
3. **Refit the article's models** (logit, MRQAP, ERGM, and the LSM if time allows) on each treated network and pool with Rubin's rules.
4. **Compare with the complete-data estimates of Section 2:** bias and RMSE of each coefficient, coverage of the complete-data value by the pooled 95% interval, sign and significance changes for the contrasts in Table 1, and whether the ranking of the models by BIC and goodness of fit changes.

The relevant derived covariate needs care. *Alter's influence in-degree* is computed from the influence-attribution network, which is itself a survey layer. When a respondent is missing, their nominations are also missing, so the covariate is recomputed from each completed network rather than taken from the full data. Otherwise the true values would leak into the imputation. The benchmark already enforces this rule.

Timeline. Benchmark long run: November 2026. Extension runs on the Swiss network: alongside, in R, with the `relimpute` imputations exported to R. Final course paper: December 2026.

4 Repository

All data and the R code that produce every figure and table in this document are in the folder `Final Project` of https://github.com/kadircihanduran/PLSC-508. `README.txt` in that folder, and `README.md` at the top of the repository, describe the role of each file. In short: `replicate_Cranmer2017.R` is the one script that produces everything; `check_lsm_transposed.R` tests the explanation of the LSM difference; `data/` holds the seven raw CSV files and the codebook from the replication archive; `results/` holds every figure and table in this document and the two fitted models; `compat/` installs the package versions the authors used into a separate folder; `report/` holds the LaTeX source and the PDF of this document.

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